

Zadanie 3 z listy 7

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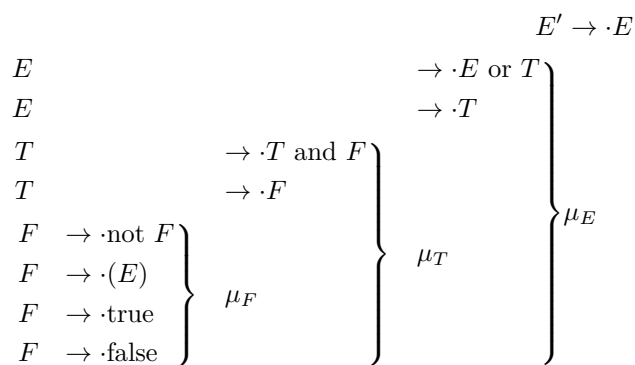
Gramatyka

Rozważana gramatyka (z rozszerzonym symbolem startowym E'):

1. $E' \rightarrow E$
2. $E \rightarrow E \text{ or } T$
3. $E \rightarrow T$
4. $T \rightarrow T \text{ and } F$
5. $T \rightarrow F$
6. $F \rightarrow \text{not } F$
7. $F \rightarrow (E)$
8. $F \rightarrow \text{true}$
9. $F \rightarrow \text{false}$

Zbiory sytuacji i przejścia goto:

I_0 :



$goto(I_0, E) = I_1$
 $E' \rightarrow E \cdot$
 $E \rightarrow E \cdot \text{or} T$

$goto(I_0, T) = I_2$
 $E \rightarrow T \cdot$
 $T \rightarrow T \cdot \text{and} F$

$goto(I_0, F) = I_3$
 $T \rightarrow F \cdot$

$goto(I_0, \text{not}) = I_4$
 $F \rightarrow \text{not} \cdot F$
 $F \rightarrow \cdot \text{not} F$
 $F \rightarrow \cdot (E)$
 $F \rightarrow \cdot \text{true}$
 $F \rightarrow \cdot \text{false}$

$goto(I_0, ()) = I_5$
 $F \rightarrow (\cdot E)$
 μ_E

$goto(I_0, \text{true}) = I_6$
 $F \rightarrow \text{true} \cdot$

$goto(I_0, \text{false}) = I_7$
 $F \rightarrow \text{false} \cdot$

$goto(I_1, \text{or}) = I_8$
 $E \rightarrow E \text{or} \cdot T$
 μ_T

$goto(I_2, \text{and}) = I_9$
 $T \rightarrow T \text{and} \cdot F$

$goto(I_4, F) = I_{10}$
 $F \rightarrow \text{not} F \cdot$

$goto(I_4, \text{not}) : I_4$
 $goto(I_4, ()) : I_5$
 $goto(I_4, \text{true}) : I_6$
 $goto(I_4, \text{false}) : I_7$
 $goto(I_5, E) = I_{11}$
 $F \rightarrow (E \cdot)$
 $E \rightarrow E \cdot \text{or} T$

$goto(I_5, T) : I_2$
 $goto(I_5, F) : I_3$
 $goto(I_5, \text{not}) : I_4$
 $goto(I_5, () : I_5$
 $goto(I_5, \text{true}) : I_6$
 $goto(I_5, \text{false}) : I_7$
 $goto(I_8, T) = I_{12}$
 $E \rightarrow \text{or}T \cdot$
 $T \rightarrow T \cdot \text{and}F$

$goto(I_8, F) : I_3$
 $goto(I_8, \text{not}) : I_4$
 $goto(I_8, () : I_5$
 $goto(I_8, \text{true}) : I_6$
 $goto(I_8, \text{false}) : I_7$
 $goto(I_9, F) = I_{13}$
 $T \rightarrow T \text{and}F \cdot$

$goto(I_9, \text{not}) : I_4$
 $goto(I_9, () : I_5$
 $goto(I_9, \text{true}) : I_6$
 $goto(I_9, \text{false}) : I_7$
 $goto(I_{11},)) = I_{14}$
 $F \rightarrow (E) \cdot$

$goto(I_{11}, \text{or}) : I_8$
 $goto(I_{12}, \text{and}) : I_9$

Zbiory FIRST

$$FIRST(F) = \{\text{not}, (, \text{true}, \text{false}\}$$

$$FIRST(T) = FIRST(F)$$

$$FIRST(E) = FIRST(T)$$

Zbiory FOLLOW

$$FOLLOW(E) = \{\$,), \text{or}\}$$

$$FOLLOW(T) = \{\$,), \text{or}, \text{and}\}$$

$$FOLLOW(F) = \{\$,), \text{or}, \text{and}\}$$

Tablica SLR

Terminale: or, and, not, (,), true, false, \$

Nieterminale: E, T, F

Stan	not	(	true	false	)	or	and	\$	E	T	F
0	s4	s5	s6	s7					1	2	3
1						s8		acc			
2					r3	r3	s9	r3			
3					r5	r5	r5	r5			
4	s4	s5	s6	s7					11		10
5	s4	s5	s6	s7						2	3
6					r8	r8	r8	r8			
7					r9	r9	r9	r9			
8	s4	s5	s6	s7						2	3
9	s4	s5	s6	s7							13
10					r6	r6	r6	r6			
11					s14	s8					
12					r2	r2	s9	r2			
13					r4	r4	r4	r4			
14					r7	r7	r7	r7			